PRACTICAL: IMPLEMENTATION OF K-MEANS ALGORITHM

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Aim

To implement the K-Means clustering algorithm using Python and visualize the resulting clusters.

Python Program

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

# Create dataset
data = np.array([
    [20, 20000], [22, 22000],
    [25, 25000], [30, 30000],
    [35, 35000], [40, 40000],
    [45, 45000], [50, 50000],
    [55, 55000], [60, 60000]
])

print("Original Dataset:")
print(data)

# Standardize data
scaler = StandardScaler()
scaled_data = scaler.fit_transform(data)

# Apply K-Means
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
kmeans.fit(scaled_data)
labels = kmeans.labels_

print("\nCluster Assignments:")
for i in range(len(data)):
    print(data[i], "Cluster:", labels[i])

# Display centroids
centers = scaler.inverse_transform(kmeans.cluster_centers_)
print("\nCluster Centers:")
print(centers)

# Visualize clusters
plt.scatter(data[:, 0], data[:, 1], c=labels, cmap='viridis')
plt.scatter(centers[:, 0], centers[:, 1],
            color='red', marker='X', s=200)
plt.xlabel("Age")
plt.ylabel("Income")
plt.title("K-Means Clustering")
plt.show()

print("K-Means Completed Successfully!")
```

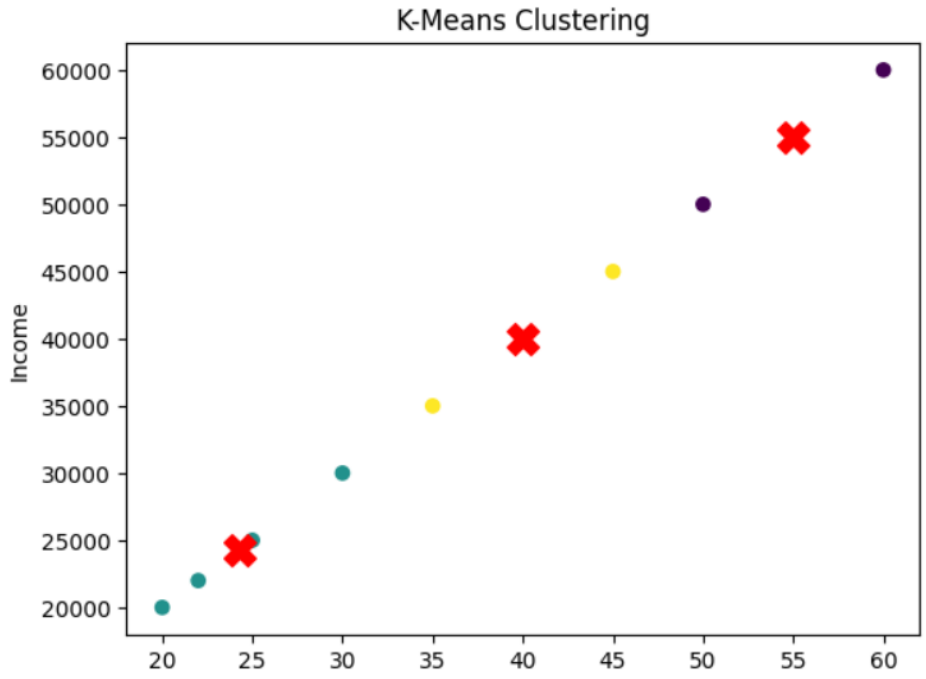
Output

The program displays the original dataset, cluster assignments, cluster centres, and a scatter plot. Different colours represent the three clusters, and red X marks represent the centroids.

```
Original Dataset:
[[   20 20000]
 [   22 22000]
 [   25 25000]
 [   30 30000]
 [   35 35000]
 [   40 40000]
 [   45 45000]
 [   50 50000]
 [   55 55000]
 [   60 60000]]

Cluster Assignments:
[   20 20000] Cluster: 1
[   22 22000] Cluster: 1
[   25 25000] Cluster: 1
[   30 30000] Cluster: 1
[   35 35000] Cluster: 2
[   40 40000] Cluster: 2
[   45 45000] Cluster: 2
[   50 50000] Cluster: 0
[   55 55000] Cluster: 0
[   60 60000] Cluster: 0
```

```
Cluster Centers:
[[5.500e+01 5.500e+04]
 [2.425e+01 2.425e+04]
 [4.000e+01 4.000e+04]]
```



Result

Thus, the K-Means clustering algorithm was successfully implemented using Python. The dataset was divided into three clusters, cluster labels and centroids were obtained, and the results were visualized using a scatter plot.